

Pre-requisites

Description

The **RISING WATER** flood prediction system was developed using a collection of Python-based software tools, libraries, and frameworks that support data preprocessing, machine learning model development, visualization, and web application deployment. These technologies provide an efficient environment for building, training, testing, and deploying the flood prediction model. The following software and libraries are required to successfully develop, execute, and maintain the project.

1. Anaconda Navigator

Description:

Anaconda Navigator is a desktop graphical user interface used to manage Python environments, packages, and data science applications. It simplifies package installation and dependency management, making it easier to configure the development environment for machine learning projects.

Website: <https://www.anaconda.com/download>

2. PyCharm

Description:

PyCharm is a professional Integrated Development Environment (IDE) for Python programming. It provides features such as intelligent code completion, debugging, syntax highlighting, project management, and version control integration, enabling efficient software development.

Website: <https://www.jetbrains.com/pycharm/>

3. Python

Description:

Python is the primary programming language used to develop the RISING WATER flood prediction system. It is widely used for machine learning, data analysis, and web application development because of its simplicity, extensive libraries, and strong community support.

Website: <https://www.python.org/>

4. NumPy

Description:

NumPy is a powerful numerical computing library that provides support for multidimensional arrays, matrices, and mathematical operations. It is used for efficient numerical calculations and preprocessing of weather data before training the machine learning model.

Website: <https://numpy.org/doc/stable/>

5. Pandas

Description:

Pandas is a data manipulation and analysis library that provides flexible data structures such as DataFrames. It is used for importing datasets, cleaning data, handling missing values, filtering records, and preparing weather data for machine learning.

Website: <https://pandas.pydata.org/docs/>

6. Scikit-learn

Description:

Scikit-learn is a popular machine learning library that provides various supervised and unsupervised learning algorithms. It is used to train, test, and evaluate flood prediction models using algorithms such as Random Forest, Decision Tree, Logistic Regression, and Support Vector Machine.

Website: <https://scikit-learn.org/stable/>

7. Matplotlib

Description:

Matplotlib is a visualization library used to generate graphs, charts, and plots. It helps analyze weather data, visualize prediction results, and evaluate the performance of machine learning models.

Website: <https://matplotlib.org/stable/>

8. Seaborn

Description:

Seaborn is a statistical data visualization library built on top of Matplotlib. It is used to create informative and attractive visualizations such as heatmaps, distribution plots, correlation matrices, and feature comparison graphs.

Website: <https://seaborn.pydata.org/>

9. Flask

Description:

Flask is a lightweight Python web framework used to develop the web interface of the RISING WATER system. It enables users to submit weather data, interact with the machine learning model, and view flood prediction results through a web browser.

Website: <https://flask.palletsprojects.com/>

10. Visual Studio Code (Optional)

Description:

Visual Studio Code is a lightweight source code editor that supports Python development through extensions. It provides debugging, syntax highlighting, integrated terminal support, and Git integration for efficient coding.

Website: <https://code.visualstudio.com/>

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum (8 GB recommended)
- Hard Disk: 10 GB free storage
- Operating System: Windows 10/11, Linux, or macOS

Software Requirements

- Python 3.10 or above
- Anaconda Navigator
- PyCharm IDE
- Flask Framework
- NumPy
- Pandas
- Scikit-learn
- Matplotlib
- Seaborn
- Web Browser (Google Chrome, Microsoft Edge, or Mozilla Firefox)

Conclusion

The above software tools and libraries provide the necessary environment for developing, training, testing, and deploying the **RISING WATER** flood prediction system. Together, they enable efficient data processing, machine learning model development, visualization of analytical results, and deployment of the application through a user-friendly web interface.